[METRIC]

DOD-D-51467(EA)
25 February 1980
(See section 6)

MILITARY SPECIFICATION

DECONTAMINATING SOLUTIONS (METRIC)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of decontaminating solutions.

1.2 Classification. Decontaminating solutions shall be of the following types as specified (see 6.2):

Type I - Decontaminating Solution I
Type II - Decontaminating Solution II

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 4230

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 5-77-2366 - Decontaminating Kit, Skin, M258A1

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. All materials cited on drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.2 Decontaminating Solution I.

3.2.1 1-Hydroxyethane (ethanol) (Drawing B5-77-2264). The ethanol content shall be 72 ± 2 percent by weight when tested as specified in 4.2.4.1.

3.2.2 Phenol (Drawing B5-77-2263). The phenol content shall be 10.0 ± 0.5 percent by weight when tested as specified in 4.2.4.2.

3.2.3 Sodium hydroxide (Drawing B5-77-2260). The sodium hydroxide content shall be 5.0 ± 0.5 percent by weight when tested as specified in 4.2.4.4.

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3.2.4 Ammonia (Drawing B5-77-2261). The ammonia content shall be 0.20 ± 0.05 percent by weight when tested as specified in 4.2.4.3.

3.2.5 Water (Drawing B5-77-2267). The balance of Decontaminating Solution I shall be made up of water conforming to the requirements of Drawing B5-77-2267.

3.3 Decontaminating Solution II.

3.3.1 1-Hydroxyethane (ethanol) (Drawing B5-77-2264). The ethanol content shall be 45 ± 2 percent by weight when tested as specified in 4.2.4.1.

3.3.2 Zinc chloride (Drawing B5-77-2265). The zinc chloride content shall be 5.0 ± 0.5 percent by weight when tested as specified in 4.2.4.5.

3.3.3 Water (Drawing B5-77-2267). The balance of Decontaminating Solution II shall be made up of water conforming to the requirements of Drawing B5-77-2267.

3.3.4 pH. The pH of Decontaminating Solution II shall be adjusted to between 4.0 and 5.2 using hydrochloric acid conforming to Drawing B5-77-2268 when tested as specified in 4.2.4.6.

3.4 Storage. The decontaminating solutions shall be stored in tightly closed, filled polyvinyl chloride, polyethylene, or nonreactive glass containers labeled to indicate solution number, date of manufacture, and lot number.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the quantity of each solution produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions and subjected to a single final mixing or blending operation in one tank.

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4.2.2 Sampling for test. A sample of no less than 100 milliliters (ml) shall be taken from each lot of solution. The sample shall be taken while the solution is contained in the final mixing tank.

4.2.3 Inspection procedure. Each sample shall be tested in accordance with the applicable portions of 4.2.4. Failure of any analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Ethanol content.

(a) Response factor. Initially determine the ethanol response factor by running gas chromatograms on a series of known mixtures of ethanol and an internal standard (methyl ethyl ketone). Calculate the response factor as follows:

$$\text{Response factor} = \frac{AB}{CD}$$

where: A = Known weight of ethanol,
B = Peak area for internal standard,
C = Known weight of internal standard, and
D = Peak area for ethanol.

(b) Procedure. To determine the concentration of ethanol in the sample, run the gas chromatogram using the same instrument and conditions as used in the response factor determination. Compare the peak area with that of the known internal standard and calculate the percent by weight ethanol as follows:

$$\text{Percent ethanol} = \frac{100REF}{GH}$$

where: E = Peak area for ethanol,
F = Weight of internal standard,
G = Peak area for internal standard,
H = Weight of sample, and
R = Response factor calculated above.

4.2.4.2 Phenol content. Weigh approximately 0.2 gram (g) of sample to the nearest 0.1 milligram (mg) in a weighing bottle. Quantitatively transfer this sample to a 500-ml iodine flask containing 25.0 ml of potassium bromate-potassium bromide solution (made by adding 3.5 g of

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potassium bromate and 13.0 g of potassium bromide in a liter of water). Add 4 ml of concentrated hydrochloric acid. Insert the stopper and swirl to mix. Let sit for 5 minutes. Add 10 ml of 10-percent potassium iodide solution quickly. Swirl to mix and wash the sides of the flask. Titrate the iodine released with 0.1N standardized sodium thiosulfate solution to the colorless end point. Starch indicator may be added near the end of the titration. Run a blank determination (water instead of sample) using the same volume of reagents. Calculate the percent by weight phenol as follows:

$$\text{Percent phenol} = \frac{1.568(B - A)C}{D}$$

where: A = Milliliters of thiosulfate solution used to titrate sample,
B = Milliliters of thiosulfate solution used to titrate blank,
C = Normality of thiosulfate solution, and
D = Weight of sample in grams.

4.2.4.3 Ammonia content. Weigh a 5 to 6 g sample to the nearest milligram into a distillation flask containing 50 ml of water. Add 10 ml of 20 percent aqueous sodium hydroxide. In a beaker or flask place 25 ml of 4 percent aqueous boric acid plus 4 drops of 0.1 percent methyl red. Make sure that the tail piece on the condenser is below the surface of the boric acid. Immediately connect the distillation flask to the still head and heat to distill 30 ml of the contents of the flask. Remove the tail piece and rinse with water into the collection flask. Titrate the contents of the flask to a red color with 0.1N hydrochloric acid. Calculate the percent by weight ammonia as follows:

$$\text{Percent ammonia} = \frac{1.705AB}{C}$$

where: A = Milliliters of hydrochloric acid used in titration,
B = Normality of the hydrochloric acid, and
C = Weight of sample in grams.

4.2.4.4 Sodium hydroxide content. Weigh a 0.5 g sample to the nearest 0.1 mg in a weighing bottle. Transfer the sample to a flask containing 20 ml of water and 5 drops of 0.1 percent alcoholic methyl red. Titrate to the brick red color of the indicator with 0.1N hydrochloric acid. Calculate the percent by weight sodium hydroxide as follows:

$$\text{Percent sodium hydroxide} = 4\left(\frac{AB}{C} - 0.59D\right)$$

where: A = Milliliters of hydrochloric acid used in the titration,
B = Normality of the hydrochloric acid,
C = Sample weight in grams, and
D = Percent ammonia determined in 4.2.4.3.

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4.2.4.5 Zinc chloride content. Weigh 5 g of sample to the nearest milligram. Dilute to 1 liter. Dilute 5 ml of this solution to 500 ml and determine the weight of zinc chloride on an atomic absorption unit using a zinc cathode lamp comparing the results to a standard curve obtained for known zinc chloride solutions run at the same time. Calculate the percent by weight zinc chloride in the solution as follows:

$$\text{Percent zinc chloride} = \frac{100AB}{W}$$

where: A = Weight of zinc chloride from calibration curve in grams,
B = Dilution factor, and
W = Weight of sample in grams.

4.2.4.6 pH. Determine the pH of the solution using a pH meter accurate to 0.1 pH units.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

6.1 Intended use. The decontaminating solutions covered by this specification are intended for use in the Decontaminating Kits, Skin, M258 and M258A1.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of decontaminating solution required (see 1.2).

6.3 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Description EA-D-1149, dated 17 February 1976.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4230-A057

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